Supplementary Information for: Helium Step Heats on Geothite at Water Saturated Conditions Show Diffusive Helium Loss and High Retentivity at Surface Temperatures

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S1. Permeation Lag Correction and Ampoule Permeability

S1.1. Permeation Lag Correction

Helium evolved from silica glass encapsulated Win-06-01b goethite sample was only available for measurement after a secondary process of permeation through the silica glass ampoule, complicating the analysis. This effect is tractable, however. Following the procedures outlined in Thyer et al. (2026), we made sequences of isothermal low-temperature measurements ($\leq 115°C$) for which the sample was assumed not to have evolved helium between each high-temperature step where the sample does evolve significant helium. The low-temperature steps, refer temperature experimental steps in order to correct for the permeation lag. However, for ^4He, the meas inventory of the ampoule between high-temperature steps.

As a result, permeation lag is instead addressed using the bounding method described in Thyer et al. (2026), which constrains the ampoule's residual helium from above rather than measuring it directly. In this approach, an upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. 7 of Thyer et al. (2026). ^{3}He was reduced this way as well, for consistency with ^{4}He. Because ^{3}He can also be reduced by the permeation step method, the two reductions agree and produce the same results. This internal test of the bounding method is reported in Thyer (2026b).

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Both the bounding and permeation step methods require the ampoule's permeation rate constant λ, which was measured (for ^{3}He) during the experiment itself following the procedure of Thyer et al. (2026).

Ten series of 3–4 consecutive permeation steps held at a single temperature between 90–110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus^{10K} were used to measure the permeation rate constant λ. The ^{3}He measurements in these series were fit to Eq. 6 of Thyer et al. (2026) using a non-linear least squares optimization (Fig. **??**). One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, although it agreed with the other measurements within uncertainty. The fitted rate constants span 2.7×10^{-5} to 6.3×10^{-5} s^{-1}, with 2σ uncertainties of 0.08 to 0.46 log units and no trend resolvable against step duration.

All measured apparent ^{3}He rate constants were equal to or in excess of those predicted by the ampouole geometry (Fig. **??**) and literature values for anhydrous silica glass (Fig. **??**), similar to observations discussed in Thyer et al. (2026). Because the presence of water does not effect the relative permeation rates of ^{3}He and ^{4}He, the ^{4}He permeability is assumed to similarly exceed the literature values for anhydrous silica glass. We therefore use the predictions by the geometry and permeabilities from **?** and **?** as a lower bound estimate of the true permeability, which translates to a conservative (higher) upper bound on the residual helium inventory of the ampoule.

The first three experimental steps had values of λt lower than 4, and therefore could not be interpreted using the bounding method and are not discussed further in this work. The permeation step method, however, can be applied to recover the ^{3}He signal and shows that these measurements extend the same smooth decline observed in the arrhenius and delta plots of the main text (Figs. 2 and 3, main text) to lower cumulative fractional yields and do not affect the results or conclusions of this work (Thyer, 2026b).

S2. Kinetic mechanisms and the limits of a single heating schedule

Solid-state rate laws separate the temperature dependence of the rate from its dependence on the extent of reaction (**?**):

$$\frac{d\alpha}{dt} = A \exp\left(-\frac{E_a}{RT}\right) h(\alpha), \tag{S1}$$

where α is the fraction of goethite converted. The reaction model $h(\alpha)$ expresses the mechanism: it describes how the rate changes as conversion

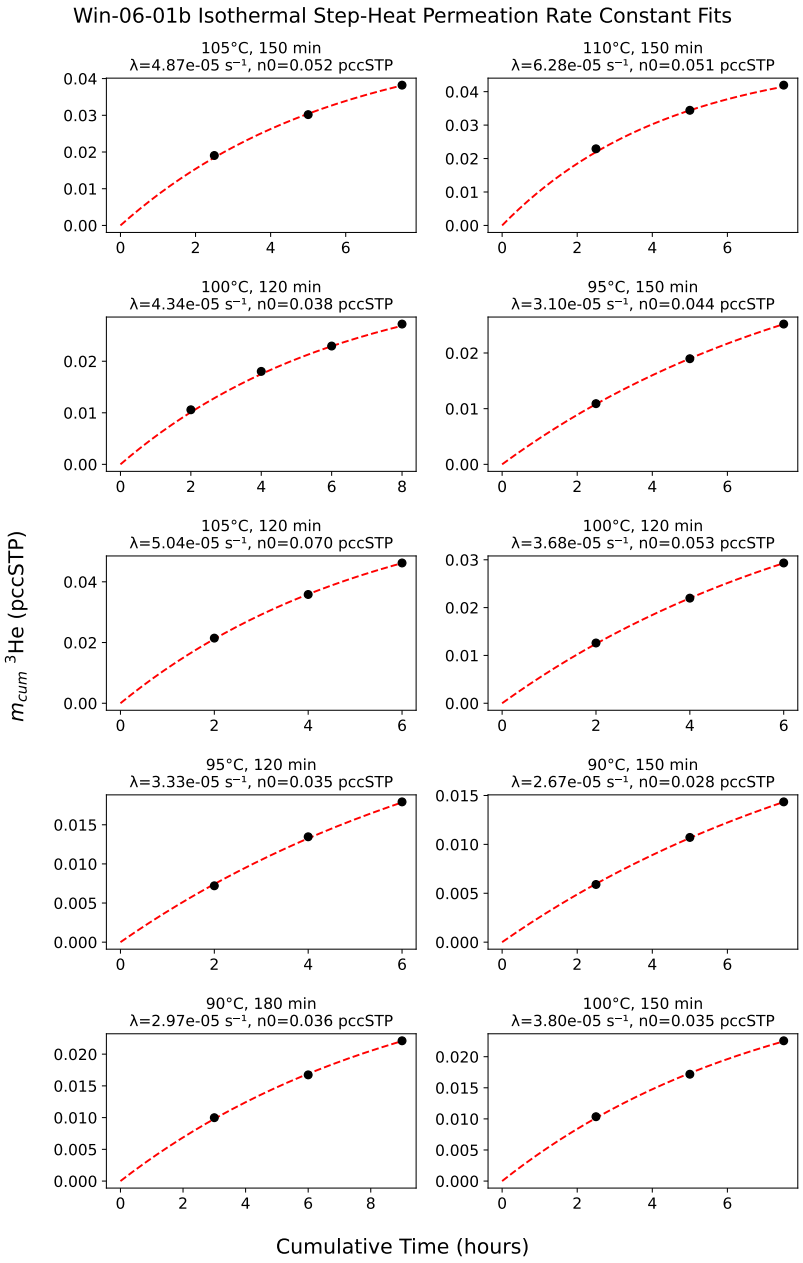


Figure S1: Cumulative yield against cumulative time for helium yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 (the rate constant of helium permeation and the initial Helium inventory of the ampoule) to these data using Eq. 6 of Thyer et al. (2026). These measured rate constants are plotted in Fig. **??**.

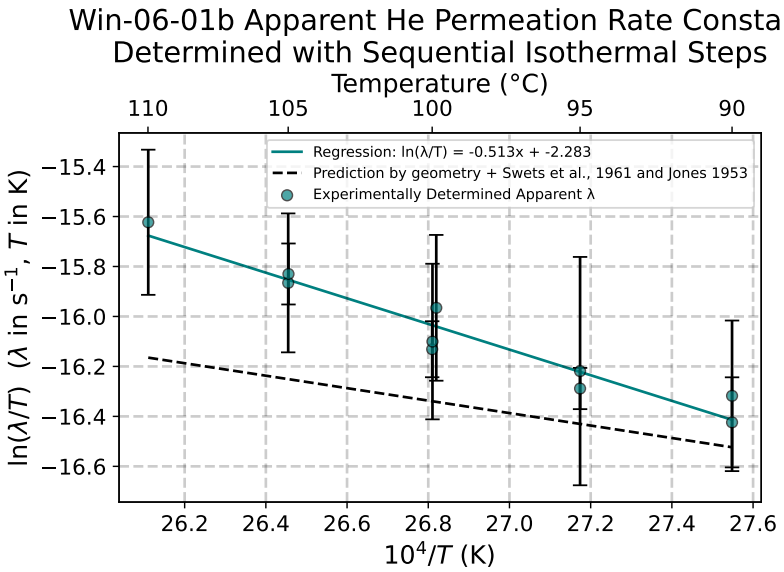


Figure S2: Arrhenius plot of the measured permeation rate constants as $\ln(\lambda/T)$ against inverse temperature for HT-Win-06-01b. The dashed line is the prediction from the ampoule geometry and the literature permeabilities of **?** and **?**.

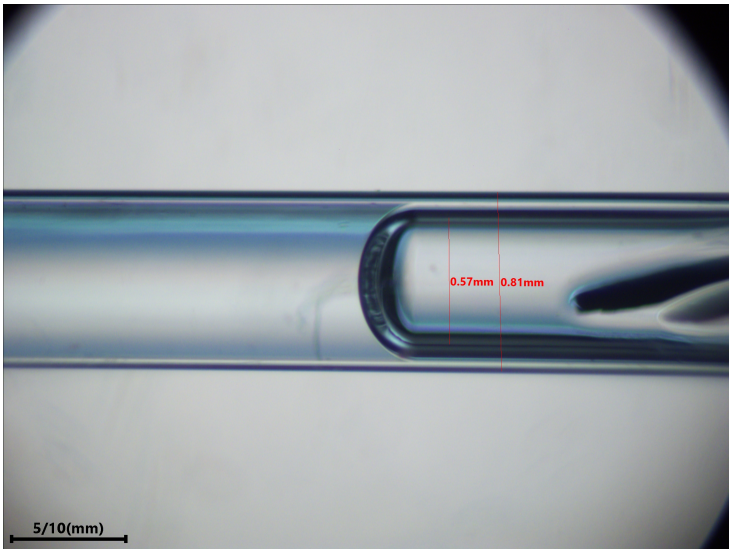


Figure S3: Measurements of the inner and outer diameter of the ampoule used in the water vapor saturated step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.



Figure S4: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

proceeds, which depends on where and how the reaction advances through a grain. For any heating schedule, each rate law gives α as a function of a single variable,

$$x = A \sum_j \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right), \tag{S2}$$

which is the cumulative Arrhenius-weighted time. When fitting the helium release of the KAF and HDM vacuum experiments, we tested the mechanisms given in Table **??**, including spherical diffusion.

Table S1: Kinetic models: reaction model $h(\alpha)$ and conversion as a function of the Arrhenius-weighted time, $\alpha(x)$.

Model	$h(\alpha)$	$\alpha(x)$
R2 contracting cylinder	$2(1-\alpha)^{1/2}$	$1-(1-x)^2$
R3 contracting sphere	$3(1-\alpha)^{2/3}$	$1-(1-x)^3$
F1 first order	$1-\alpha$	$1-e^{-x}$
A2, A3 Avrami ($n=2,3$)	$n(1-\alpha)[-\ln(1-\alpha)]^{1-1/n}$	$1-e^{-x^n}$
D3 sphere diffusion	—	$1-\frac{6}{\pi^2}\sum_{m=1}^{\infty}\frac{e^{-m^2\pi^2 x}}{m^2}$

S2.1. Fits to the vacuum data

Table **??** gives the fit of each model to the cumulative KAF ^{3}He release within the fitting window. It also gives the misfit of the resulting out-of-sample prediction of HDM. The simple first order model F1 gives the best fit and the best prediction. Spherical diffusion with $\phi = 0$ fits KAF 4.4 times worse than F1 and cannot produce the $F \approx 0.89$ plateau. To compare diffusion on the same footing, we also fit D3 with ϕ fixed at the F1 value and with ϕ free, over the same 18 KAF steps. The imposed plateau improves the KAF misfit to 0.026 and 0.019but the out-of-sample prediction of HDM does not improve (rms 0.056 and 0.052, against 0.018 for F1).

The fitting window excludes the low-temperature steps, where release is dominated by diffusion from goethite, and the plateau, where it is dominated by diffusion from hematite. Repeating the fits with every step included gives the same result: F1 remains the best model ($E_a = 122.7\,\mathrm{kJ\,mol^{-1}}$, $\phi = 0.112$, rms 0.0073 for KAF and 0.026 for the HDM prediction.

Table S2: Kinetic and Diffusive Model fits to the vacuum experiments (from Monteiro (2017) and Farley et al. (2024)). D3 is fitted with ϕ fixed at 0, fixed at the F1 value, and free.

Model	E_a	$\ln A$	ϕ	rms KAF	rms HDM
R2	92.8	10.63	0.149	0.0133	0.0291
R3	103.1	12.73	0.136	0.0114	0.0240
F1	120.8	18.15	0.101	0.0087	0.0176
A2	25.5	−4.73	0.000	0.0212	0.1906
A3	20.0[a]	−5.85	0.203	0.0433	0.2342
D3, $\phi = 0$	195.6	32.43	0.000	0.0385	0.0484
D3, $\phi = \phi_{\mathrm{F1}}$	242.8	43.91	0.101	0.0258	0.0557
D3, ϕ free	288.5	55.04	0.184	0.0186	0.0520

E_a in $\mathrm{kJ\,mol^{-1}}$, A in $\mathrm{s^{-1}}$. [a]At the lower bound of the fit.

S3. Thermodynamics of Goethite Dehydroxylation

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \;\rightleftharpoons\; \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\text{fluid})}, \tag{S3}$$

whose Gibbs energy change, written per mole of hematite formed, is

$$\Delta G_r(T,P) = G_{\mathrm{hem}}(T,P) + G_{\mathrm{w}}(T,P) - 2\,G_{\mathrm{goe}}(T,P) + \Delta G_{\mathrm{surf}}. \tag{S4}$$

where ΔG_r is the Gibbs energy change of the reaction and G_{hem}, G_{w}, and G_{goe} are the Gibbs energies of hematite, water, and goethite respectively. ΔG_{surf} is the surface energy contribution, which must be considered to reproduce observed phase stability (Navrotsky et al., 2008). Goethite is stable where $\Delta G_r > 0$. The values and equations used to evaluate each of these terms are given below.

Gibbs energies of the solids. The Gibbs energy of a phase at (T, P) follows from integrating $dG = -S\,dT + V\,dP$ from the reference state $(T_r, P_r) = (298.15\ \mathrm{K}, 1\ \mathrm{bar})$, first in temperature at P_r and then in pressure at T. At the reference pressure, $G(T,P_r) = H(T) - TS(T)$ where $H(T) = \Delta_f H_i^0 + \int_{T_r}^{T} C_p\,dT$ and $S(T) = S_i^0 + \int_{T_r}^{T} (C_p/T)\,dT$. Adding $\int_{P_r}^{P} V\,dP$, and treating the solids as incompressible with no thermal expansion gives

$$G_i(T,P) = \Delta_f H_i^0 + \int_{T_r}^{T} C_{p,i}\,dT \;-\; T\left[S_i^0 + \int_{T_r}^{T} \frac{C_{p,i}}{T}\,dT\right] \;+\; V_i\,(P - P_r), \tag{S5}$$

This is the "apparent" Gibbs energy of formation (**?**); the element-entropy terms that distinguish it from the true Gibbs energy of formation cancel in any balanced reaction. Compressibility and thermal expansion change V_i by $\lesssim 1\%$ over the conditions considered and are neglected.

Goethite heat capacity uses the Berman–Brown form (**?**) refined by Majzlan et al. (2003), $C_p = k_0 + k_1 T^{-1/2} + k_2 T^{-2} + k_3 T^{-3}$. Hematite uses the same polynomial with the coefficients of **?**, which Majzlan et al. (2003) held fixed in their refinement. Both heat capacities are integrated analytically.

Water. Water is evaluated with the IAPWS-95 equation of state (**?**), implemented in CoolProp (**?**), which is valid for liquid, vapor and supercritical water to 1 GPa. In the same apparent convention,

$$G_{\mathrm{w}}(T,P) = \Delta_f H^0_{\mathrm{w}} + [h(T,P) - h(T_r,P_r)] - T\left[S^0_{\mathrm{w}} + s(T,P) - s(T_r,P_r)\right], \tag{S6}$$

where h and s are the IAPWS-95 molar enthalpy and entropy and the reference state is liquid water at T_r, P_r. Table **??** lists all input data. We use the enthalpies and entropies of Majzlan et al. (2003).

Table S3: Thermodynamic data used for Eq. (**??**).

	Goethite (α-FeOOH)	Hematite (α-Fe_2O_3)
$\Delta_f H^0$ ($\mathrm{kJ\,mol^{-1}}$)	-561.4	-825.8
S^0 ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	59.5	87.4
V^0 ($\mathrm{cm^3\,mol^{-1}}$)	20.881	30.274
C_p coefficients	$k_0 = 133.8765$	$k_0 = 146.86$
	$k_1 = -738.5577$	$k_1 = 0$
	$k_2 = -2.177657 \times 10^6$	$k_2 = -5.5768 \times 10^6$
	$k_3 = 2.165124 \times 10^8$	$k_3 = 5.2563 \times 10^8$
$C_p(298\ \mathrm{K})$ ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	74.8	104.0
$\gamma_{\mathrm{hydrated}}$ ($\mathrm{J\,m^{-2}}$)	0.60 ± 0.10	0.75 ± 0.16
$\gamma_{\mathrm{anhydrous}}$ ($\mathrm{J\,m^{-2}}$)	0.91 ± 0.09	1.90 ± 0.30

Enthalpy, entropy, goethite volume and C_p: Majzlan et al. (2003); hematite volume and C_p: **?**; surface energies: Navrotsky et al. (2008). Water: $\Delta_f H^0_{\mathrm{w}} = -285.83\ \mathrm{kJ\,mol^{-1}}$, $S^0_{\mathrm{w}} = 69.95\ \mathrm{J\,mol^{-1}\,K^{-1}}$ (**?**).

Surface energy. A crystal of finite size carries an excess free energy at its surface in addition to the free energy of its bulk (**?**). For a population of particles of fixed size, this excess is γA, where γ is the surface energy ($\mathrm{J\,m^{-2}}$) and A the total surface area. At fixed particle size, both the bulk and surface terms scale with the amount of material, so the surface contribution can be written per mole, where A_m is the surface area per mole. For the reaction as

written (one mole of hematite from two moles of goethite),

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}} A_{m,\mathrm{hem}} - 2\,\gamma_{\mathrm{goe}} A_{m,\mathrm{goe}}, \tag{S7}$$

where $A_{m,\mathrm{hem}}$ is per mole of Fe_2O_3 and $A_{m,\mathrm{goe}}$ per mole of FeOOH.

A size must be chosen for each phase. Following Navrotsky et al. (2008), we assume pseudomorphic dehydroxylation, in which hematite inherits the external envelope of the parent goethite, so that both phases share the same external surface area per mole of Fe (allowing us to collapse $A_{m,hem}$ and $A_{m,goe}$ to A_{Fe}). For spherical goethite grains of diameter d this shared area is

$$A_{\mathrm{Fe}} = \frac{6\,V_{\mathrm{goe}}}{d} \quad \text{per mole of Fe.} \tag{S8}$$

Goethite has one Fe per formula unit, so $A_{m,\mathrm{goe}} = A_{\mathrm{Fe}}$. Hematite has two, so $A_{m,\mathrm{hem}} = 2A_{\mathrm{Fe}}$. Substituting into Eq. (**??**),

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}}\,(2A_{\mathrm{Fe}}) - 2\,\gamma_{\mathrm{goe}}\,A_{\mathrm{Fe}} = 2A_{\mathrm{Fe}}\,(\gamma_{\mathrm{hem}} - \gamma_{\mathrm{goe}})\,. \tag{S9}$$

The factor of two counts the two moles of Fe that pass through the reaction.

We use hydrated surface energies. They are referenced to liquid water, so they apply wherever the surfaces are in equilibrium with water at or near saturation, which includes all of the water vapor saturated experiments. Anhydrous surface energies apply only to surfaces that have lost their chemisorbed water, i.e. under vacuum.

Phase diagram.

S3.1. Stability and driving force

The phase diagram (Fig. **??**) is computed by solving $\Delta G_r(T, P) = 0$ (Eqs. (**??**)–(**??**)) numerically for $P = P_{\mathrm{H_2O}}$ at each temperature using Brent's method. It communicates several important features of the goethite–hematite–water system that are relevant to the experiments considered here:

1. Goethite is not stable at the conditions of any of the experiments considered here, in water or in vacuum, in spite of the evidence that it did not react in the water-saturated experiment. Thus, we hypothesize that the reaction is kinetically inhibited for goethite under these conditions.
2. The stability of goethite is strongly dependent on the partial pressure of water below and up to water saturation. Above the saturation pressure, water is liquid and contributes far less to the reaction volume than it does as a gas, becoming a secondary control on goethite's stability.

3. The stability of goethite is strongly dependent on grain size, with smaller grains being more stable than larger grains. At water saturation, bulk goethite is stable below 18 °C, 100 nm goethite below 28 °C, and 10 nm goethite below 108 °C.

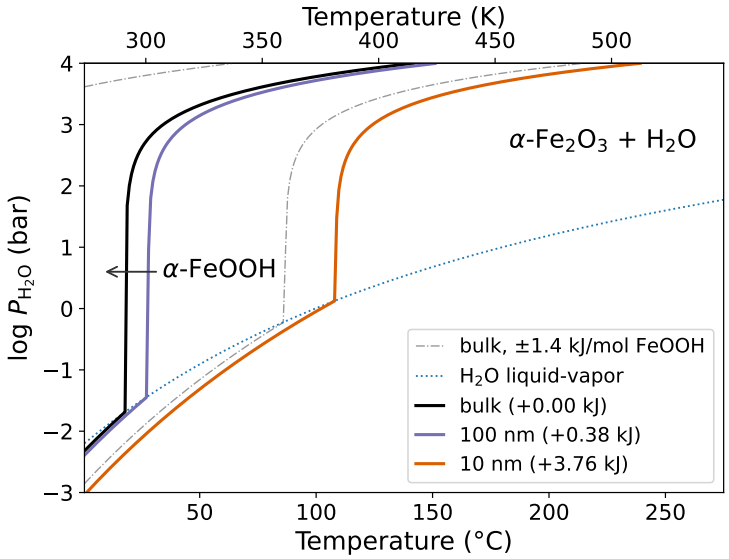


Figure S5: Goethite–hematite–water equilibrium in $\log P_{H_2O}$–T space for bulk goethite and for 100 and 10 nm grains with hydrated surfaces, compared at equal surface area per mole of Fe (Eq. (**??**)). Goethite is stable above and to the left of each curve. The dash-dot curves illustrate the absolute uncertainty: it is the bulk stability boundary shifted by the ±1.4 kJ mol^{-1} FeOOH uncertainty of Majzlan et al. (2003). The blue dotted curve shows the water vapor saturation pressure; liquid water is present at and above this curve. The thermodynamic data used to compute the phase diagram are in Table **??**.

Stability alone does not determine what happens in a step heat; the rate of conversion controls whether this equilibrium state is reached or if the reaction proceeds at all, and that depends on the chemical potential of the reaction: $\Delta\mu$. Table **??** gives the bulk driving forces at the experimental temperatures, evaluated with IAPWS-95 water. At saturation, $\Delta\mu$ is 6–11 kJ, only 1.6–2.4 times RT. Under the vacuum of a step-heating line the water pressure approaches zero, where $\Delta\mu$ is unbounded (arbitrarily large). The chemical potential of water is therefore the key variable separating the two experimental settings.

Table S4: Bulk driving force for hematite formation, $\Delta\mu = -\Delta G_r$ (kJ per mole Fe_2O_3), at water saturation. In vacuum ($P_{H_2O} \to 0$) the driving force is unbounded.

T (°C)	P_{sat} (bar)	$\Delta\mu_{sat}$	$\Delta\mu_{sat}/RT$
150	4.76	5.68	1.61
175	8.93	6.88	1.84
200	15.55	8.11	2.06
225	25.50	9.37	2.26
250	39.76	10.66	2.45

S4. Nucleation Model of Goethite Dehydroxylation

Nucleation barrier. In classical nucleation theory (**???**), reviewed by **?**, **?** and, for mineral systems, **?** and **?**, a spherical hematite nucleus of radius r has free energy

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \frac{\Delta\mu}{V_{hem}} + 4\pi r^2\gamma, \tag{S10}$$

where $\Delta\mu$ is the driving force per mole of Fe_2O_3 (Table **??**), V_{hem} the molar volume and γ the energy of the interface surrounding the nucleus. Nuclei larger than the critical radius $r^* = 2\gamma V_{hem}/\Delta\mu$ grow, and the energetic barrier to forming one is

$$\Delta G^* = \frac{16\pi\,\gamma^3\,V_{hem}^2}{3\,\Delta\mu^2}. \tag{S11}$$

In the first-order limit of §4.1, the rate constant is a nucleation rate of the form derived by Turnbull and Fisher (1949),

$$k = A\exp\!\left(-\frac{E_a}{RT}\right)\exp\!\left(-\frac{\Delta G^*}{k_BT}\right),$$
$$\frac{\Delta G^*}{k_BT} = \frac{C\gamma^3}{T\,\Delta\mu^2}, \qquad C \equiv \frac{16\pi V_{hem}^2}{3k_B} = 1.112\times10^{15}\ \mathrm{m^6\,K\,J^{-1}\,mol^{-2}}, \tag{S12}$$

where $A\exp(-E_a/RT)$ describes the atomic mobility and $\exp(-\Delta G^*/k_BT)$ is the equilibrium fraction of sites occupied by critical nuclei. Because the barrier scales as $\gamma^3/\Delta\mu^2$, it vanishes in vacuum where $\Delta\mu$ goes to infinity, but it is significant at water saturation, where $\Delta\mu$ is only 6–11 kJ (Table **??**).

Internal nucleation. Goethite dehydroxylates topotactically: hematite nucleates within the parent crystal, inherits its oxygen framework and orientation, and the pseudomorph develops pores and twinned hematite domains

(????). We therefore consider only internal nucleation, in which the new interface of the burgeoning nuclei is between hematite and goethite. We know of no measured or calculated energy for that interface, so γ is replaced by an effective parameter γ_{eff}, calibrated from the data. *Calibration.* When Farley et al. (2024) isolated the liquid nitrogen trap from the heated goethite, they raised $P_{\text{H}_2\text{O}}$ in the line and lowered $\Delta\mu$, and the isothermal rate fell by a factor s (§3.2). With A and E_a unchanged, Eq. (??) gives

$$\gamma_{\text{eff}} = \left[\frac{T\,\Delta\mu_{\text{off}}^2 \ln(1/s)}{C}\right]^{1/3}, \tag{S13}$$

where $\Delta\mu_{\text{off}}$ is evaluated at an upper bound on the water pressure with the trap isolated. This pressure estimate of the water released depends on the volume of the experimental apparatus and the water content emitted by the goethite, which was measured with capacitance manometry With the trap in place, the water pressure approaches zero, the driving force is unbounded, and its contribution to the barrier vanishes. Any finite water pressure in the trapped line would give a larger γ_{eff}, so this limit is conservative. This gives $\gamma_{\text{eff}} = 0.14\ \text{J m}^{-2}$ (range 0.12–0.15). At saturation and 250 °C, the nucleation barrier is 32–63 k_BT: the critical nucleus has a radius of 0.7–0.9 nm and a volume of 26–52 formula units. For nuclei this small, classical nucleation theory has no literal meaning, so γ_{eff} is best read as the single parameter of a rate law in which $\ln k$ falls as $-1/\Delta\mu^2$, not as an interface energy.

Translation to water saturation. For the same sample, only $\Delta\mu$ changes between vacuum and saturation:

$$\frac{k_{\text{sat}}}{k_{\text{vac}}} = \exp\left[-\frac{C\gamma_{\text{eff}}^3}{T}\frac{1}{\Delta\mu_{\text{sat}}^2}\right]. \tag{S14}$$

Applied step by step to the water vapor saturated schedule, Eq. (??) predicts a total conversion ($\sum_i \frac{k_{\text{sat}}}{k_{\text{vac}}}(T_i)\,k_{\text{vac}}(T_i)\,\Delta t_i$) of $10^{-12.4}$, $10^{-20.7}$ and $10^{-25.8}$ for $\gamma_{\text{eff}} = 0.12$, 0.14 and 0.15 J m^{-2}, and $10^{-38.7}$ to $10^{-9.1}$ (95%) when the uncertainties in the calibration and in the thermodynamic data are propagated (§??). The exact values reflect the assumed $\Delta\mu^{-2}$ dependence; what matters is that the model predicts suppression far beyond the $10^{-3.8}$ that the data require (main text, §4.3).

Assumptions. This is a consistent physical explanation of the data, not a proof. It is not exhaustive. This analysis assumes (i) that the trap-isolated slowdown acts through $\Delta\mu$, rather than, for example, through water blocking

reactive sites; (ii) that γ_{eff}, A and E_a are the same in vacuum and in water; (iii) that the barrier depends on the bulk driving force without significant contribution from surface terms (this analysis extended to 10 nm grains raises the predicted conversion only to $\sim 10^{-13}$; §**??**); and (iv) that conversion is solid-state. Conversion by dissolution of goethite and precipitation of hematite from the fluid is not described by Eq. (**??**) (see §**??**). The calibrated γ_{eff} absorbs contributions from the sample's microstructure and cannot be transferred to other goethites.

S4.1. Calibration of γ_{eff}

Equation (**??**) follows from Eq. (**??**) applied to a trap-isolated step and its trapped neighbors, $\ln(1/s) = C\gamma_{\text{eff}}^3/(T\,\Delta\mu_{\text{off}}^2)$; the corresponding term for the trapped neighbors vanishes because their driving force is unbounded (item 2). The driving forces are evaluated as follows.

1. $\Delta\mu_{\text{off}}$ is a bulk driving force, $-\Delta G_r$ from Eq. (**??**) (§**??**) with $\Delta G_{\text{surf}} = 0$, at the step temperature and with the solids at vacuum.
2. With the trap in place, the water pressure in the line approaches zero, so the chemical potential of water falls without bound, $\Delta\mu_{\text{on}}$ is unbounded, and its term $\Delta\mu_{\text{on}}^{-2}$ vanishes. This is the conservative limit: any finite water pressure in the trapped line would give a larger γ_{eff} and a stronger predicted slowdown at saturation. $\Delta\mu_{\text{off}}$ is evaluated at the water pressure reached with the trap isolated (below).
3. Surface terms are omitted. The barrier of an internal nucleus depends on the free energy per volume of material transformed, not on the external surface of the host grain. The external surface energy of the grain enters the overall thermodynamic stability (§**??**), but is not a part of the nucleation barrier, which is evaluated for internal nuclei.

We bound the water pressure with the trap isolated by assuming that all evolved water was in the vapor, distributed uniformly through an apparatus of volume V_{app} at 298 K:

$$P_{\text{H}_2\text{O}} = \frac{(m_{\text{w}}/M_{\text{w}})\,RT}{V_{\text{app}}}. \tag{S15}$$

With $m_{\text{w}} = 0.027$ mg at 230 °C and 0.016 mg at 240 °C (1.50 and 0.89 μmol), and $190 \le V_{\text{app}} \le 600$ mL, Table **??** gives the pressures and the resulting γ_{eff}. The pressures are end-of-step values; we also evaluate half of each as

a time average for steady accumulation. This is an upper bound on the pressure, because some water adsorbs on the walls of the line; a lower true pressure gives a larger γ_{eff} and more suppression at saturation, so the bound is conservative. Pressure enters only through $\ln P$ in $\Delta\mu_{\text{off}}$ and then through $\Delta\mu_{\text{off}}^{2/3}$, so a threefold uncertainty in volume changes γ_{eff} by only about 5%. The two steps agree within 20%, and we adopt $\gamma_{\text{eff}} = 0.14\ \text{J m}^{-2}$, with a range of 0.12–0.15 J m^{-2}.

Table S5: Water pressure with the trap isolated and the calibrated γ_{eff} (J m^{-2}; Eq. (**??**)).

T (°C)	s	V_{app} (mL)	$P_{\text{H}_2\text{O}}$ (Pa)	γ_{eff}, end-of-step P	γ_{eff}, $P/2$
230	0.333	190	19.6	0.120	0.124
230	0.333	600	6.2	0.126	0.130
240	0.224	190	11.6	0.140	0.145
240	0.224	600	3.7	0.147	0.152

S4.2. Uncertainty propagation

The Monte Carlo range quoted in §4.3 (4000 draws) samples the trap-isolated step (230 or 240 °C); s, log-uniformly between 0.30–0.37 at 230 °C and 0.22–0.23 at 240 °C; the apparatus volume, log-uniformly between 190 and 600 mL; a time-averaging coefficient for the pressure between 0.5 and 1 to account for how quickly the evolved water accumulates in the diffusion cell, and a common offset of ΔG_r, uniformly within ±2.8 kJ per reaction (the ±1.4 kJ per mole FeOOH of Majzlan et al. 2003), which shifts every driving force together. The offset barely changes γ_{eff} (95% range 0.12–0.15 J m^{-2}) but contributes most of the spread in the predicted conversion, because it enters the small $\Delta\mu_{\text{sat}}$ (6–11 kJ) squared.

S5. Multiple Diffusion Domain Fits and Thermal History Results

The modeling procedure is described in the main text (§2.2) and follows Farley and Flowers (2012) and Shuster and Farley (2004).

S5.1. Multiple diffusion domain fits for $N = 1$–5

Figure **??** shows the fitted fractional and cumulative ^{3}He yields, and the Arrhenius and delta plots they imply, for one to five domains. The fits for all N are summarized in Table **??**.

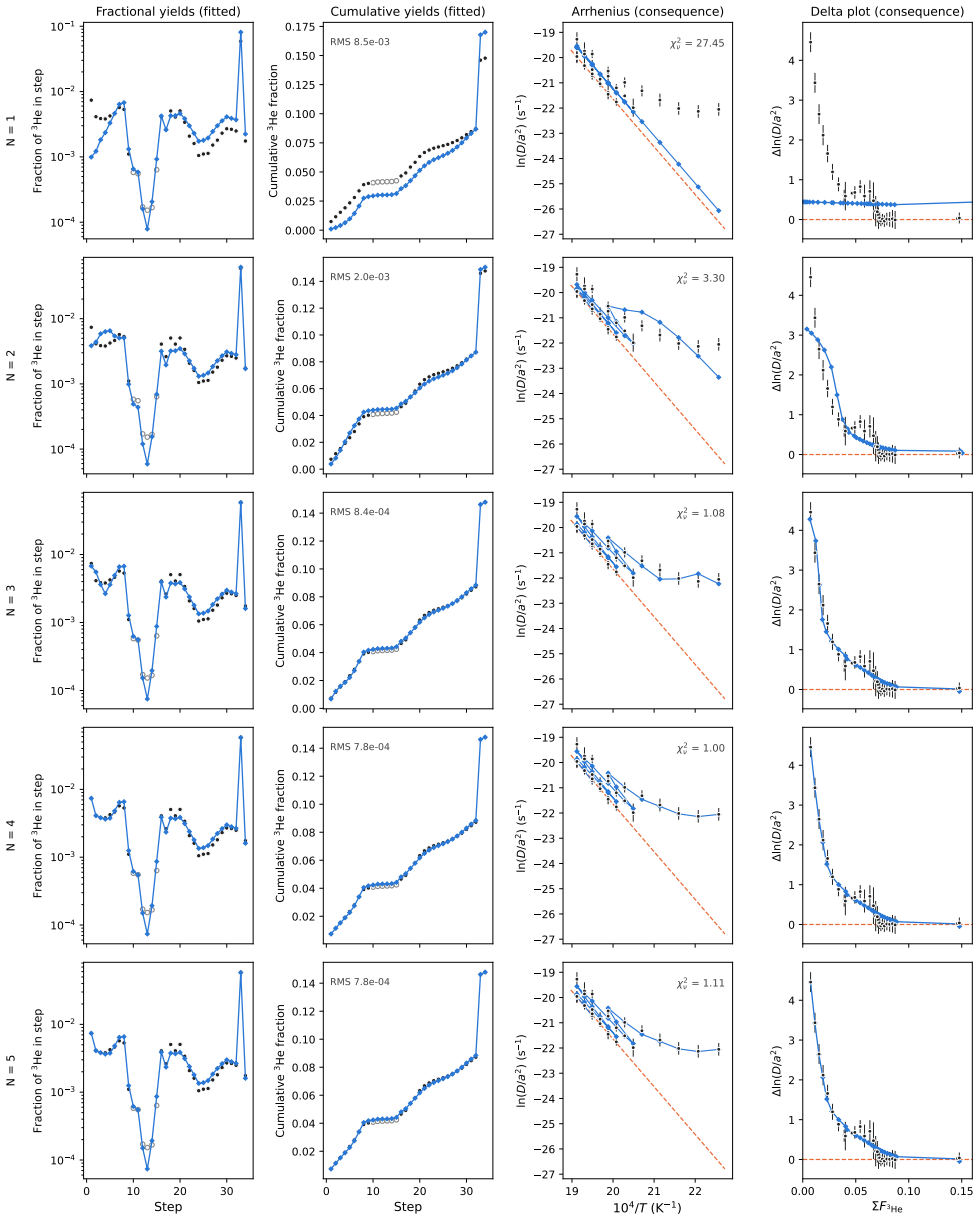


Figure S6: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b for $N = 1$–5 domains (rows). Columns: fitted fractional and cumulative yields, and the Arrhenius and delta plots implied by each fit. Hollow grey symbols are the six low-signal steps, which are not fitted and are omitted from the Arrhenius and delta plots.

Table S6: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b (N domains, $k = 2N - 1$ parameters). RMS is the yield misfit ($\times 10^{-4}$, fraction of total ^{3}He) over the 28 steps above the signal threshold. "Fast" is the ^{3}He fraction in domains with $r/r_0 < 0.1$; f_{main} and r_{main}/r_0 are the fraction and size of the largest-fraction domain. For $N \geq 5$, the dominant domain is split among several domains of identical size.

N	k	RMS	Arrh. χ^2_ν	fast (%)	f_{main} (%)	r_{main}/r_0
1	1	85.30	27.45	0.0	100.0	0.80
2	3	20.20	3.30	2.2	97.8	1.05
3	5	8.45	1.08	2.7	97.3	1.12
4	7	7.81	1.00	2.8	97.2	1.12
5	9	7.81	1.11	2.8	95.1	1.12
6	11	7.81	1.24	2.8	85.6	1.12
7	13	7.81	1.40	2.8	85.3	1.12
8	15	7.81	1.62	2.8	76.8	1.12
9	17	7.81	1.91	2.8	76.5	1.12
10	19	7.81	2.34	2.8	74.6	1.12

Table S7: Thermal histories fitted to the ^{4}He/^{3}He spectrum of HT-Win-06-01b with the four-domain structure fixed. Each row gives the best path with a predicted He age of 130–180 Ma along with the reduced χ^2 value ($\nu = 22$ for steady cooling and the isothermal hold, $\nu = 21$ for burial). For steady cooling, $T_{\mathrm{max}} = T_0$ is reached at crystallization, so $t_{\mathrm{peak}} = t_{\mathrm{form}}$. For the isothermal hold, $T_{\mathrm{max}} = T_{\mathrm{iso}}$ throughout. Burial crystallization ages are fixed within each row. The last three columns give the ranges of the age-consistent paths within the joint 1σ region about the best fit (main text, §2.2).

Path	t_{form} (Ma)	t_{peak} (Ma)	T_{max} (°C)	χ^2_ν	t_{He} (Ma)	t_{form} (1σ)	t_{peak} (1σ)	T_{max} (1σ)
Linear cooling	150	150	78	2.00	132	150–175	150–175	76–80
Burial	180	70	61	5.36	164	fixed	60–82	58–65
Burial	195	80	61	5.04	177	fixed	68–98	58–65
Burial	210	115	71	6.53	180	fixed	105–130	70–74
Isothermal hold	150	–	41	18.5	141	140–190	–	39–43

S6. Fluid-Mediated Conversion of Synthetic Goethite Pigment

S6.1. Methods

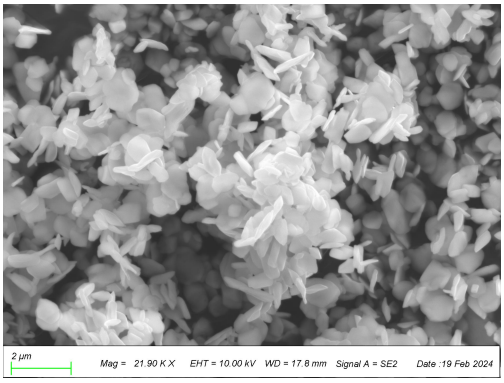
Synthetic goethite pigment, a commercial yellow iron-oxide pigment of acicular crystals ∼1 µm long, was treated in the same way as the Bahia goethite (main text, §2.3) but only with water, without vacuum or open-air runs: at least five larger silica glass ampoules, each loaded with pigment and an excess of water and fused shut at ambient pressure, were held at 200–250 °C for about a day in an oven whose voltage was adjusted by hand. These runs were a rapid, qualitative test; details of individual runs (temperature histories, durations, water-to-solid ratios) were not recorded, and reaction was assessed by color (yellow goethite, red hematite) before imaging.

S6.2. Results and Interpretation

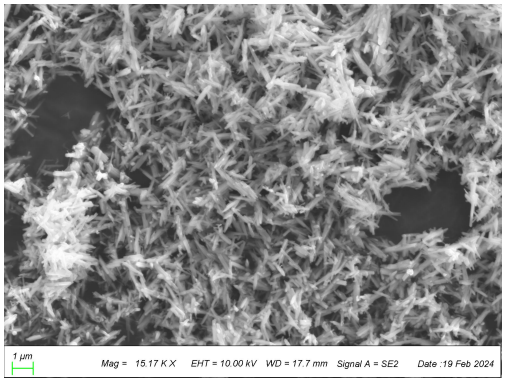
We observed the spontaneous conversion of synthetic goethite pigment (§**??**) to hematite when heated in ampoules with excess water at experimental temperatures. Roughly equal proportions of ampoules were completely reacted (red), completely unreacted (yellow), or contained both reacted and unreacted material in isolated patches. In the mixed ampoules, the red and yellow portions were entirely isolated from one another, separated by gas bubbles. Unreacted pigment retains its original acicular habit (Fig. **??**b). The reacted hematite product is instead composed of equant to platy crystals ∼0.5–1 µm across (Fig. **??**a). The change of shape rules out pseudomorphic dehydroxylation like that observed to occur in goethite heated in vacuum (Fig. 1), indicating dissolution of goethite and nucleation and growth of hematite from the fluid, a pathway outside the solid-state nucleation model developed in §4.3.

Conversion either went to completion within a fluid-connected volume or did not occur at all, even where reacted and unreacted pigment shared the same ampoule and thermal history. We posit that the fluid-mediated reaction requires some threshold of dissolved Fe(III) in the fluid to nucleate hematite, but because the reaction consumes and produces dissolved Fe(III) in equal proportions, it is not self-limiting and proceeds quickly to completion once that threshold is reached.

Grain size and bulk thermodynamics cannot explain why the pigment blades reacted when the natural polycrystalline goethite did not. At 1 µm, the surface contribution to ΔG_r is only ∼0.04 kJ, and the bulk driving force at saturation is the same 6–11 kJ for both materials (Table **??**). The most



(a) Reacted (red) product



(b) Unreacted (yellow) pigment

Figure S7: Secondary electron images of synthetic goethite pigment after treatment at water vapor saturation and 200–250 °C for about a day (§**??**). (a) Material from a completely reacted (red) ampoule: the acicular goethite has been replaced by equant to platy hematite crystals ∼0.5–1 µm across. (b) Material from an unreacted (yellow) ampoule, which retains the unchanged acicular habit of the starting pigment. The change in crystal habit requires dissolution of goethite and precipitation of hematite from the fluid, and the all-or-nothing outcome of the experiments indicates that the reaction is fast once it begins.

likely difference is the fluid itself. Commercial iron-oxide pigments are commonly produced by precipitation from iron-salt solutions. They can carry residual soluble salts such as iron and alkali sulfates or nitrates, and they are often surface-treated with organometallic compounds (**???**). Such contaminants would change the pH and ionic strength of the water they are heated in, and with them the solubility of Fe(III). The dissolution–precipitation pathway is limited by how much iron the fluid carries, so a modest increase in Fe(III) solubility could make a nucleation-limited reaction viable. The natural Win-06-01b goethite, by contrast, was heated in distilled water, where near-neutral Fe(III) solubility is very low, potentially offering an explanation for the divergent outcomes.

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